

Patentability Search Report

On

"A System and Method for Risk-Based Stateful Control of AI Agent Interactions for Preventing Instruction Manipulation and Data Leakage"

Report Details

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PATENTABILITY SEARCH REPORT

1. Objective

To assess the patentability of the proposed invention aimed at the development of a risk-based stateful control system for securing Large Language Model (LLM)-based AI agents against prompt injection, instruction manipulation, and context leakage during interactive operations, under the provisions of the Indian Patents Act, 1970, specifically evaluating novelty under Section 2(1)(l) read with Section 13, inventive step under Section 2(1)(ja), industrial applicability under Section 2(1)(j), and exclusions under Section 3.

2. Understanding of the Proposed Subject Matter for Search

As noted from the invention disclosure provided, the proposed invention relates to a computer-implemented control system for securing Large Language Model (LLM)-based AI agents against prompt injection, instruction manipulation, and context leakage during interactive operations. The system is configured to regulate AI agent behavior through continuous risk evaluation, explicit state transitions, and coordinated multi-layer enforcement, enabling stable, adaptive, and context-aware protection across multi-turn interactions.

The system addresses the gap in existing AI security mechanisms where security enforcement is treated as a post-processing or filtering task rather than as a continuous control problem governing AI agent behavior across the entire interaction lifecycle. Existing solutions do not implement explicit computational state representations with defined transition conditions, resulting in uniform responses irrespective of evolving threat levels. When adversarial interactions are not captured and systematically analyzed through stateful control mechanisms, AI agents may be susceptible to staged manipulation patterns, context leakage, and instruction override attacks.

Key aspects include:

- Input Processing and Analysis Module: The system accepts user prompts, system inputs, and interaction events from AI agent interfaces in real-time, performs semantic analysis, context extraction, and instruction parsing to identify input intent and structure, executes tool invocation checks, and logs interaction attributes for risk evaluation and behavioral tracking.
- Unified Multi-Factor Risk Computation Engine: A centralized risk evaluation mechanism computes a continuous risk score (R) using a weighted multi-parameter

function defined over normalized feature vectors incorporating semantic deviation scores, instruction consistency gradients, context sensitivity coefficients, and temporal interaction weights, aggregating current input signals with historical interaction data to produce a normalized risk value.

→ Stateful Control and Decision Module: A formally defined state machine governs AI agent behavior, wherein system states (including SAFE, SUSPICIOUS, UNDER ATTACK, and LOCKDOWN) are dynamically updated based on computed risk scores, temporal interaction patterns, and feedback signals, enabling context-dependent and condition-aware response strategies.

→ Dynamic Context Exposure Regulation Module: A control mechanism selectively restricts or permits access to internal system prompts, memory representations, and contextual data based on real-time risk scores and system state, applying selective masking, truncation, or isolation of contextual information under elevated risk conditions to prevent unauthorized disclosure of sensitive information.

→ Multi-Layer Security Enforcement Module: Coordinated security actions are enforced across input filtering and sanitization, reasoning control and constraint enforcement, tool execution regulation, and output validation and filtering layers under a unified control policy derived from risk and state conditions.

→ Feedback-Driven Adaptation Mechanism: The system continuously monitors outputs and interaction outcomes, adjusts risk computation parameters and state transition thresholds based on observed behavior, and incorporates feedback signals to improve decision stability and reduce false positives.

→ Attack Memory and Pattern Learning Module: An integrated memory subsystem stores historical adversarial interaction patterns and detected attack signatures, utilizes stored patterns to enhance risk prediction and early detection of recurring threats, and enables proactive identification of staged and multi-turn attack strategies.

Thus, based on the above understanding of the subject matter and information provided, we have conducted a preliminary patentability search and have identified a few documents which may be relevant to the proposed subject matter. Same are listed below under Section 4 of the report for your review/comments.

3. Search Strategy

Relevant Keywords and Synonyms:

Keywords	Concept/Synonyms
Prompt injection defense	Prompt attack prevention, injection detection, adversarial prompt filtering
LLM security	Large language model protection, AI agent security, foundation model defense
Stateful control	State machine, security state transitions, state-based enforcement
Risk scoring	Risk computation, threat assessment, security scoring, risk evaluation
Context leakage prevention	Data leakage protection, context exposure control, memory protection
Multi-turn interaction	Sequential dialogue, conversational security, multi-round analysis
Attack memory	Adversarial pattern learning, attack pattern storage, threat memory
AI agent guardrails	Safety guardrails, security enforcement, behavioral constraints

Search Queries:

No.	Search Query
1.	("prompt injection" OR "adversarial prompt" OR "instruction manipulation") AND ("LLM" OR "large language model" OR "AI agent") AND ("defense" OR "protection" OR "security")
2.	("stateful control" OR "state machine" OR "security state") AND ("AI" OR "language model") AND ("risk" OR "threat" OR "attack")
3.	("context leakage" OR "data leakage" OR "memory protection") AND ("LLM" OR "AI agent") AND ("prevention" OR "control")
4.	("multi-turn" OR "multi-round" OR "sequential interaction") AND ("attack detection" OR "adversarial" OR "manipulation") AND ("language model")
5.	("risk scoring" OR "risk computation" OR "threat assessment") AND ("AI security" OR "LLM" OR "agent") AND ("real-time" OR "dynamic")

IPC Classes: G06F 21/55, G06F 21/56, G06F 21/57, G06N 3/08, G06F 40/30

CPC Classes: G06F 21/554, G06F 21/566, G06N 3/084, G06F 40/30, G06F 21/577

Databases used: Google Patents, Espacenet, Google Scholar, Patentscope, PQAI, Patseer, Indian Patent Advanced Search

Coverage: Worldwide and published prior art till date.

4. Identified Prior Art Documents with Their Relevant Disclosures

To assess the potential patentability of the subject invention, prior art references have been categorized based on their relevance to the claimed subject matter. The following tiered classification system has been applied to indicate the degree to which each reference relates to the inventive features:

Relevance Tier Classification:

Tier 1: References of high relevance that disclose subject matter closely aligned with the core inventive concepts of the application. These may pose significant challenges to novelty or inventive step.

Tier 2: References of moderate relevance that share certain features with the invention but do not disclose all key inventive aspects.

Relevance	Prior Art	Relevant Disclosure	Remarks / Overlapping Features / Key Differences
Tier 1	CN-121561061-A "Multi-round Dialogue Security Defense Method, Apparatus, Electronic Device, Storage Medium and Program	The invention discloses a multi-round dialogue security defense method that obtains multi-round dialogue content and tool invocation sequences, constructs a dialogue semantic graph representing semantic evolution, intent flow, and entity association, as	Key Similarities: Both inventions process multi-round dialogue content and analyze sequential interactions. Both inventions construct graph-based representations for risk analysis. Both inventions extract risk feature vectors from user inputs. Both inventions dynamically update

	Product" (China, 2025)	well as a tool invocation graph reflecting temporal and interaction relationships of multiple tool invocations. User input information is input into both graphs to extract dual risk feature vectors based on multi-round context and tool invocation logic. The system performs collaborative risk judgment using preset rules to determine defense results including intercept and release. The dialogue semantic graph and tool invocation graph are dynamically updated based on risk vectors after each interaction round.	their models based on interaction outcomes. Key Differences: Present invention implements a deterministic state machine with four explicit security states (SAFE, SUSPICIOUS, UNDER ATTACK, LOCKDOWN) with defined transition conditions, not disclosed in prior art. Present invention computes a continuous risk score using a weighted multi-parameter function incorporating semantic deviation scores, instruction consistency gradients, context sensitivity coefficients, and temporal interaction weights, whereas prior art uses graph neural network-derived feature vectors with MLP classification. Present invention includes a Dynamic Context Exposure Regulation Module that selectively restricts access to internal system prompts and memory representations based on risk and state, not disclosed in prior art. Present invention
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			implements temporal persistence wherein elevated security states are maintained despite subsequent low-risk inputs to prevent adversarial bypass, not disclosed in prior art.
Tier 1	CN-121262016-B "Defense Method, System, Terminal and Medium for LLM Prompt Attacks" (China, 2025)	The invention discloses a defense method for LLM prompt attacks that receives user input, generates multiple detection input copies based on preset rules, and inputs them in parallel into independent detection processes to obtain multiple risk scores. The system constructs a comprehensive risk score through weighted fusion using category sensitivity factors and process credibility factors, determines risk level, and when the risk level falls within a defense trigger interval, executes active defense strategies including instruction obfuscation, semantic dilution, and structural	Key Similarities: Both inventions compute comprehensive risk scores through weighted fusion of multiple factors. Both inventions execute multiple types of attack detection in parallel. Both inventions implement dynamic intervention mechanisms that adjust based on risk levels. Both inventions apply graduated defense strategies based on risk assessment. Key Differences: Present invention implements a formally defined state machine with persistent security states (SAFE, SUSPICIOUS, UNDER ATTACK, LOCKDOWN) that maintain across interactions, whereas prior art uses stateless risk level intervals evaluated per-input.

		isolation. Independent detection processes execute privilege jailbreak detection, prompt injection detection, context-induced attack detection, and chain-of-thought hijacking detection. A dynamic intervention control mechanism determines execution order and processing intensity of defense strategies.	Present invention includes sequential interaction modeling with temporal dependency analysis for detecting staged multi-turn adversarial strategies, not disclosed in prior art which operates on individual inputs. Present invention includes a Dynamic Context Exposure Regulation Module controlling access to internal system prompts and memory, not disclosed in prior art. Present invention includes an Attack Memory and Pattern Learning Module storing historical adversarial patterns for long-term learning, not disclosed in prior art.
Tier 2	CN-121561909-A "LLM Attack Defense Method, Apparatus, Device and Storage Medium" (China, 2025)	The invention discloses an LLM attack defense method that obtains input text, performs malicious instruction risk assessment to obtain a risk assessment result, performs consistency risk assessment based on the LLM's output text and an auxiliary verification model's verification output text to obtain an	Key Similarities: Both inventions perform weighted fusion of multiple risk factors to determine risk levels. Both inventions execute graded defense strategies based on risk assessment (allow, sanitize, restrict, block). Both inventions use multi-factor risk evaluation including semantic features. Key Differences: Present

		output consistency result. Based on the risk assessment result and output consistency result, the system performs weighted linear fusion to determine the risk level of the input text and executes corresponding graded defense strategies including: response output for low risk, detection result correction output for medium-low risk, adaptive input correction output with semantic reconstruction for medium-high risk, and request blocking output for high risk.	invention implements a deterministic state machine with defined security states and temporal persistence across interactions, whereas prior art evaluates each input independently without maintaining persistent system state. Present invention includes sequential interaction modeling with temporal dependency analysis for multi-turn attack detection, not disclosed in prior art. Present invention includes a Dynamic Context Exposure Regulation Module, not disclosed in prior art. Present invention includes an Attack Memory and Pattern Learning Module for continuous defensive improvement, not disclosed in prior art. Present invention implements cross-layer feedback mechanism integrating input processing, reasoning control, tool execution, and output filtering, not disclosed in prior art.
Tier 2	CN-121765054-A "Multi-round Dialogue Risk	The invention discloses a multi-round dialogue risk detection method that	Key Similarities: Both inventions analyze multi-round dialogue content across

	Detection Method, Apparatus, Electronic Device and Storage Medium" (China, 2025)	obtains at least two rounds of dialogue information between a user terminal and an LLM via a proxy gateway, generates a multi-round dialogue snapshot based on dialogue content and dialogue turns, obtains resolved text by performing entity extraction and pronoun/ellipsis resolution, identifies instruction types and content types in the resolved text using a preset classification model, and determines the dialogue risk level based on the instruction type and content type using a preset risk level lookup table. When dialogue risk level is high risk, the system blocks the dialogue; when medium risk, it generates refusal content; when low risk, it records the dialogue information.	sequential interactions. Both inventions determine risk levels and execute graduated responses (block, refuse, record). Both inventions process dialogue content to extract structured information. Key Differences: Present invention computes a continuous risk score using a weighted multi-parameter function with four specific factors (semantic deviation, instruction consistency, context sensitivity, temporal weights), whereas prior art uses a static lookup table mapping instruction type and content type to risk levels. Present invention implements a deterministic state machine with persistent security states and temporal persistence, not disclosed in prior art. Present invention includes a Dynamic Context Exposure Regulation Module, not disclosed in prior art. Present invention includes an Attack Memory and Pattern Learning Module with feedback-driven adaptation, not disclosed in prior art.
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			Present invention implements cross-layer enforcement across input, reasoning, tool execution, and output stages, not disclosed in prior art.
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5. Overall Comments

With respect to novelty and inventive step:

Novelty: Under Section 2(1)(l) read with Section 13 of the Indian Patents Act, 1970

Based on the preliminary patentability search and review of the relevant prior art documents listed under Section 4 of this report, novelty is an absolute requirement: an invention is new only if no single prior art reference discloses all its claimed features in combination. As established in patent jurisprudence, novelty should always be ascertained with reference to a single prior art reference, and various documents cannot be combined; every element of the alleged claim should be disclosed in the prior art for novelty destroying.

The present invention is novel because no single prior art reference discloses the claimed subject matter in its entirety. While Tier 1 prior art such as CN-121561061-A teaches multi-round dialogue security defense with dual graph modeling (dialogue semantic graph and tool invocation graph), dynamic graph updates based on risk vectors, and collaborative risk judgment, it does not disclose:

- Deterministic state machine with four explicit security states (SAFE, SUSPICIOUS, UNDER ATTACK, LOCKDOWN) with defined transition conditions based on computed risk scores and temporal interaction patterns, enabling context-dependent and condition-aware response strategies.
- Unified multi-factor risk computation engine computing a continuous risk score using a weighted multi-parameter function incorporating semantic deviation scores, instruction consistency gradients, context sensitivity coefficients, and temporal interaction weights, with dynamic adjustment through feedback signals.
- Dynamic Context Exposure Regulation Module that selectively restricts or permits access to internal system prompts, memory representations, and contextual data based

on real-time risk scores and system state, applying selective masking, truncation, or isolation under elevated risk conditions.

- Temporal persistence mechanism wherein elevated security states (particularly LOCKDOWN) are maintained despite subsequent low-risk inputs, with gradual safe inputs enabling controlled recovery, preventing adversarial bypass through staged interactions.
- Closed-loop integration of risk computation, state transition control, and context exposure regulation operating as a unified event-driven control pipeline for real-time adaptive enforcement.

Similarly, Tier 1 prior art CN-121262016-B discloses parallel multi-type attack detection with weighted risk fusion and dynamic intervention control but lacks the deterministic state machine with temporal persistence, multi-turn sequential interaction modeling, dynamic context exposure regulation, and attack memory with pattern learning.

Tier 2 prior arts CN-121561909-A and CN-121765054-A disclose various aspects of risk assessment and graded defense strategies but do not anticipate the combination of stateful control with temporal persistence, unified multi-factor risk computation with four specific parameters, dynamic context exposure regulation, attack memory and pattern learning, and closed-loop cross-layer feedback integration.

Since none of the cited prior art references anticipates all essential features of the invention in combination, the subject matter is new within the meaning of Section 2(1)(l) read with Section 13 of the Indian Patents Act, 1970.

Inventive Step: Under Section 2(1)(ja) of the Indian Patents Act, 1970

Section 2(1)(ja) defines inventive step as a feature that entails a "technical advance as compared to existing knowledge or having economic significance" or both, and that makes the invention not obvious to a person skilled in the art. The determination of obviousness is done objectively, avoiding hindsight.

Furthermore, it is important to note that a subject matter will be regarded as patentable by the Indian Patent Office only if it satisfies both the novelty requirement and the inventive-step requirement under Section 2(1)(ja) of the Patents Act, 1970.

The invention satisfies the inventive step requirement under Section 2(1)(ja) because it demonstrates a significant technical advance over existing knowledge. Prior arts individually

disclose multi-round dialogue analysis, risk scoring, or attack detection, but they do not suggest or motivate the skilled person to combine these elements into:

1. Deterministic State Machine with Temporal Persistence: The system implements a formally defined state machine governing AI agent behavior with four explicit security states (SAFE, SUSPICIOUS, UNDER ATTACK, LOCKDOWN), wherein state transitions are governed by both computed risk scores AND historical interaction memory, and wherein the system maintains elevated security states despite subsequent low-risk inputs to prevent adversarial bypass. This feature provides a significant technical advancement by enabling persistent security enforcement that cannot be circumvented through staged low-risk interactions following an attack attempt. No prior art teaches or suggests this temporal persistence mechanism.

2. Unified Multi-Factor Risk Computation with Four Specific Parameters: The system computes a continuous risk score using a weighted multi-parameter function incorporating semantic deviation scores, instruction consistency gradients, context sensitivity coefficients, and temporal interaction weights, with dynamic adjustment of the weighting function through feedback signals derived from prior interaction outcomes. This represents a non-obvious solution to the problem of accurately quantifying adversarial risk across multiple dimensions while enabling adaptive refinement based on observed system behavior.

3. Dynamic Context Exposure Regulation Module: The system selectively restricts or permits access to internal system prompts, memory representations, and contextual information based on computed risk score and active system state, applying selective masking, truncation, or isolation of contextual information under elevated risk conditions. This addresses the distinct technical problem of context leakage prevention that is not contemplated by any of the cited prior art references.

4. Closed-Loop Integration of Risk, State, and Context Control: The integration of the unified multi-factor risk computation engine with deterministic state transition control and dynamic context exposure regulation forms a tightly coupled control loop, wherein computed risk values directly influence system state transitions and context access decisions in real time. This coordinated interaction across multiple system layers produces measurable technical effects including reduction in unintended context leakage, improved consistency of security responses across multi-turn interactions, and enhanced detection of staged adversarial manipulation patterns. Such a tightly integrated, feedback-driven control architecture is not

achievable through isolated or sequential application of individual techniques found in the prior art.

Under the Teaching-Suggestion-Motivation (TSM) test, none of the four prior art references, individually or in combination, provide teaching, suggestion, or motivation for a person skilled in the art to arrive at the claimed combination of features. CN-121561061-A's architecture is fundamentally graph-neural-network-based with binary classification output, and there is no teaching or suggestion to transform its binary intercept/release paradigm into a multi-state control system with temporal persistence. CN-121262016-B's architecture is single-input-oriented with parallel detection pipelines, and there is no motivation to extend this to a stateful, multi-turn control system with persistent attack memory. The prior art references operate under fundamentally different architectural paradigms that would not lead a skilled person toward the present invention's stateful, closed-loop, context-regulating control architecture.

Section 3 Exclusions:

The relevant exclusions to check are in Section 3 of the Indian Patents Act.

Section 3(k):

Under Section 3(k) of the Indian Patents Act, "a mathematical or business method or a computer program per se or algorithms" are excluded from patentability. However, the present invention does not fall under this exclusion because it is not limited to software or algorithms in isolation. Instead, it discloses a technical system comprising:

- **Hardware components:** Input processing module, risk computation engine, state machine controller, context regulation module, multi-layer enforcement module, feedback adaptation mechanism, and attack memory storage, implemented on computing devices for real-time AI agent security enforcement.
- **Technical integration:** The system integrates input processing, semantic analysis, multi-factor risk computation, deterministic state transitions, context exposure regulation, multi-layer enforcement, and feedback-driven adaptation to achieve measurable technical improvements in AI agent security.
- **Physical-world interaction:** The system interfaces with AI agent interactions to detect adversarial manipulation patterns, prevent context leakage, and enforce security policies in real-time interactive operations.

- **Tangible technical effects:** Reduction in unintended context leakage during adversarial interactions, improved consistency and stability of security responses across multi-turn interactions, enhanced detection of staged and low-intensity adversarial manipulation patterns, and controlled and adaptive system behavior under dynamically evolving threat conditions.

The invention delivers concrete security outcomes including real-time adversarial detection, context leakage prevention, and stateful security enforcement which extend beyond mere computation. The risk computation algorithms and state machine logic are applied to achieve a technical result (securing AI agent interactions against prompt injection, instruction manipulation, and data leakage) rather than claimed in isolation.

Accordingly, the invention constitutes a technical application of software integrated with hardware for AI agent security and is patent-eligible notwithstanding Section 3(k).

With Respect to the Patentability of the Proposed Subject Matter:

In view of the above analysis, the present invention is **patentable** under Indian law, as it satisfies the requirements of:

- **Novelty [Section 2(1)(i)]:** No single prior art reference discloses all essential features in combination.
- **Inventive Step [Section 2(1)(b)]:** The invention demonstrates significant technical advance through deterministic state machine with temporal persistence, unified multi-factor risk computation with four specific parameters, dynamic context exposure regulation module, and closed-loop integration of risk, state, and context control.
- **Industrial Applicability:** The invention is clearly applicable for securing LLM-based AI agents in enterprise deployments, AI-powered customer service systems, autonomous AI agent platforms, conversational AI applications, and any interactive AI system requiring protection against adversarial manipulation and data leakage.
- **Section 3 Exclusions:** The invention does not fall under Section 3(k) exclusions.

The invention represents a technical advancement in AI agent security by enabling a risk-based stateful control framework that integrates unified multi-factor risk computation, deterministic state transitions with temporal persistence, dynamic context exposure regulation, multi-layer enforcement, and feedback-driven adaptation with attack memory. It provides significant practical utility for organizations deploying LLM-based AI agents in security-sensitive

environments, particularly where traditional static filtering or isolated guardrail mechanisms are inadequate for detecting staged adversarial manipulation patterns and preventing context leakage, and is found **patentable** under Indian Patent Law.

6. Disclaimer

The data/information was obtained from the databases listed under Part 3, covering only the published documents. Our analysis in the present report is based on such data/information. The present report provides a list of results obtained by conducting a search based on search strategy, as per our understanding of the subject matter and invention disclosure provided to us. Kindly refer to "Search Strategy" Section for more details.